

Problem Summary

The tilt of a node is : $|\text{sum}(\text{left subtree}) - \text{sum}(\text{right subtree})|$

The tilt of the whole tree is the sum of tilts of all nodes.

Return the total tilt of the binary tree.

Approach

- Step1 : Global Tilt Variable
- Step2 : DFS Returns Subtree Sum
- Step3 : Base Case
- Step4 : Calculate Left and Right Sums
- Step5 : Compute Current Node Tilt
- Step6 : Return Subtree Sum

Algorithm

- Initialize global tilt=0.
- Run DFS.
- For each node:
 - Compute left subtree sum.
 - Compute right subtree sum.
 - Add current node tilt to answer.
 - Return subtree sum.
- Return total tilt.

Time Complexity & Space Complexity

TC : $O(n)$
SC : $O(h)$

Binary Tree

DescriptionEditorialSolutionsSubmissions

563. Binary Tree Tilt

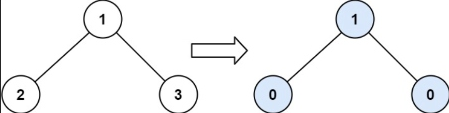
Solved

EasyTopicsCompaniesHint

Given the **root** of a binary tree, return the sum of every tree node's **tilt**.

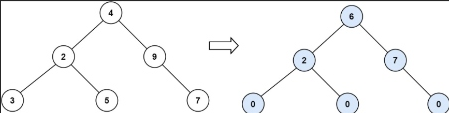
The **tilt** of a tree node is the **absolute difference** between the sum of all left subtree node **values** and all right subtree node **values**. If a node does not have a left child, then the sum of the left subtree node **values** is treated as 0. The rule is similar if the node does not have a right child.

Example 1:



Input: root = [1,2,3]
Output: 1
Explanation:
Tilt of node 2 : $|0-0| = 0$ (no children)
Tilt of node 3 : $|0-0| = 0$ (no children)
Tilt of node 1 : $|2-3| = 1$ (left subtree is just left child, so sum is 2; right subtree is just right child, so sum is 3)
Sum of every tilt : $0 + 0 + 1 = 1$

Example 2:



Input: root = [4,2,9,3,5,null,7]
Output: 4

2.4K476 Online

Accepted

All Submissions

Accepted 77 / 77 testcases passed

Sahil K... submitted at Jun 06, 2026 02:21

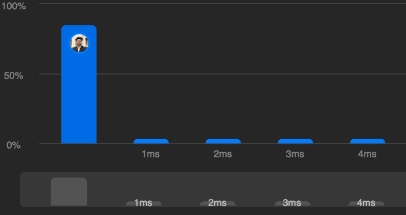
AnalysisSolution

Runtime

0 ms | Beats 100.00%

Memory

46.51 MB | Beats 28.15%



Code | Java

```
1 /**
2  * Definition for a binary tree node.
3  * public class TreeNode {
4  *     int val;
5  *     TreeNode left;
6  *     TreeNode right;
7  *     TreeNode() {}
8  *     TreeNode(int val) { this.val = val; }
9  * }
```

Write your notes here

Code

Java

```
11 *     this.left = left;
12 *     this.right = right;
13 * }
14 * }
15 */
16 class Solution {
17     int tilt = 0;
18     public int findTilt(TreeNode root) {
19         dfs(root);
20         return tilt;
21     }
22     private int dfs(TreeNode root){
23         if(root==null) return 0;
24
25         int leftSum = dfs(root.left);
26         int rightSum = dfs(root.right);
27
28         tilt += Math.abs(leftSum - rightSum);
29
30         return leftSum + rightSum + root.val;
31     }
32 }
```

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TestcaseTest Result

Accepted

Runtime: 0 ms

Case 1Case 2Case 3

Input

root = [1,2,3]

Output